

Individual Assignment 5

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1. Setup

Packages

```
# general data wrangling and plotting
library(tidyverse)

# file paths
library(here)

# clean column names
library(janitor)

# convert taxon names to snake case
library(snakecase)

# wrap long axis labels
library(scales)

# read Excel files
library(readxl)

# combine plots
library(patchwork)
```

Data

```
# read in taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# read in aquatic invertebrate data
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# read in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# read in water quality data
# na values are included so unusual entries are treated as missing
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))
```

2. Cleaning

NCOS sites

```
# create object with only NCOS quarterly sampling sites
ncos_sites <- sites |>

  # keep sites in the NCOS sector that were sampled quarterly
  filter(sector == "NCOS" &
    sampling_frequency == "Quarterly")
```

Taxon list

```
# clean taxon names so they match the aquatic invertebrate data
taxon_list_clean <- taxon_list |>

  # convert verbatim taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
    case = "snake"))
```

Water quality

```
# create cleaned water quality data frame
water_quality_clean <- water_quality |>

# clean column names
clean_names() |>

# keep only NCOS quarterly sampling sites
semi_join(ncos_sites, by = "site") |>

# create a shared date-site column for joining later
unite("date_site",
      date,
      site,
      remove = TRUE) |>

# group by each date and site combination
group_by(date_site) |>

# calculate median water quality values for each date-site survey
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>

# ungroup data frame
ungroup() |>

# remove known salinity outlier
filter(date_site != "2022-08-12_NVBR")
```

Aquatic invertebrates

```
# create cleaned aquatic invertebrate data frame
aq_ins_clean <- aq_ins |>

# clean column names
clean_names() |>

# keep only NCOS quarterly sampling sites
```

```

semi_join(ncos_sites, by = "site") |>

# rename date column for easier use
rename(date = date_on_vial) |>

# keep date, sample type, site, and major invertebrate taxon columns
select(date,
       sample_type,
       site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>

# keep only filtered beaker samples
filter(sample_type %in% c("FB 250", "FB250")) |>

# convert taxon columns into long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>

# group by date, site, and taxon
group_by(date, site, taxon) |>

# calculate average abundance per liter
summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5,
          .groups = "drop") |>

# create shared date-site column for joining with water quality
unite("date_site",
     date,
     site,
     remove = FALSE) |>

# join water quality data
left_join(water_quality_clean, by = "date_site") |>

```

```

# join taxonomic information
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>

# create full site names
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>

# reorder sites by median salinity
mutate(site_full = fct_reorder(site_full,
                              med_sal,
                              .fun = "median",
                              .na_rm = TRUE),
       site_full = fct_rev(site_full))

```

3. Class figures needed for assignment

Site salinity

```

# create salinity plot object
salinity <- ggplot(data = aq_ins_clean,

                  # x-axis: site
                  # y-axis: median salinity
                  mapping = aes(x = site_full,
                               y = med_sal)) +

# show individual salinity measurements for each survey
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +

```

```

# wrap long site names on x-axis
scale_x_discrete(labels = label_wrap(width = 15)) +

# show median salinity for each site
stat_summary(fun = median,
              geom = "point",
              color = "red",
              shape = 18,
              size = 4) +

# label axes and add title
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site salinity") +

# use theme different from default
theme_bw() +

# angle x-axis text for readability
theme(axis.text.x = element_text(angle = 45,
                                   hjust = 1))

```

Copepod abundance

```

# create data frame for copepod abundance
copepod_data <- aq_ins_clean |>

# keep only copepods
filter(taxon == "copepod") |>

# log + 1 transform average abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

# create copepod plot object
copepods <- ggplot(data = copepod_data,

                  # x-axis: site
                  # y-axis: log-transformed average abundance per liter
                  mapping = aes(x = site_full,
                                y = log_ave_lit)) +

```

```

# show individual average abundance/L measurements
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +

# show median abundance for each site
stat_summary(fun = median,
            geom = "point",
            color = "blue",
            shape = 18,
            size = 4) +

# wrap long site names on x-axis
scale_x_discrete(labels = label_wrap(width = 15)) +

# label axes and add title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Copepods") +

# use theme different from default
theme_bw() +

# angle x-axis text for readability
theme(axis.text.x = element_text(angle = 45,
                                  hjust = 1))

```

4. Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan the figure

Goal: create a plot showing differences between sites in ostracod abundance.

- x-axis: `site_full`
- y-axis: `log_ave_lit` (log + 1 transformed average abundance/L)
- geometry to represent average abundance/L: jittered points using `geom_jitter()`
- geometry to represent medians: larger colored diamond points using `stat_summary(fun = median)`

b. Wrangle data

```
# create data frame for ostracod abundance
ostracod_data <- aq_ins_clean |>

# keep only ostracods
filter(taxon == "ostracod") |>

# log + 1 transform average abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows from ostracod data frame
ostracod_data |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 24
```

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2022-08-08_NPB2	2022-08-08 00:00:00	NPB2	ostr~	2	7.68	2.36	2.43
2	2020-01-21_NDC	2020-01-21 00:00:00	NDC	ostr~	2.4	6.53	0.3	7.6
3	2022-08-10_NPB	2022-08-10 00:00:00	NPB	ostr~	2.67	8.28	8.84	9.81
4	2023-02-28_NPB	2023-02-28 00:00:00	NPB	ostr~	2.53	NA	NA	NA
5	2022-11-18_NVBR	2022-11-18 00:00:00	NVBR	ostr~	0	NA	NA	NA
6	2022-05-11_NDC	2022-05-11 00:00:00	NDC	ostr~	0	9.11	0.398	9.4
7	2023-06-06_NPB1	2023-06-06 00:00:00	NPB1	ostr~	0.533	7.48	NA	2.38
8	2024-01-28_NPB	2024-01-28 00:00:00	NPB	ostr~	3.33	7.43	3.08	6.6
9	2023-02-28_NPB1	2023-02-28 00:00:00	NPB1	ostr~	0	NA	NA	NA
10	2023-05-04_NEC	2023-05-04 00:00:00	NEC	ostr~	0.667	8.05	NA	8.55

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,  
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,  
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,  
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code figure

```
# create ostracod plot object
ostracods <- ggplot(data = ostracod_data,

# x-axis: site
```



```

# y-axis: log-transformed average abundance per liter
mapping = aes(x = site_full,
               y = log_ave_lit)) +

# show individual average abundance/L measurements
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +

# show median abundance for each site with a visually distinct point
stat_summary(fun = median,
             geom = "point",
             color = "purple",
             shape = 18,
             size = 4) +

# wrap long site labels
scale_x_discrete(labels = label_wrap(width = 15)) +

# label axes and add title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +

# use theme different from default
theme_bw()

```

d. Create a multipanel figure

```

# combine salinity, copepod, and ostracod plots
salinity / (copepods | ostracods)

```

